Cairo University

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EEG Project

Comparative Study of Inverse Solution Methods

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Two Inverse Solution Methods

Overview

This document summarizes a comprehensive comparative analysis of two mathematical methods for solving the electroencephalography (EEG) inverse problem: the Weighted Minimum Norm (WMN) method and the Sparse Solution method based on Lasso regression. The study evaluates both approaches under clean and noisy measurement conditions to determine their relative strengths, limitations, and appropriate application contexts.

Background and Significance

The EEG Inverse Problem

Electroencephalography measures electrical potentials at the scalp surface resulting from neural activity within the brain. The inverse problem involves determining which brain sources generated the observed scalp recordings. This problem is mathematically ill-posed because:

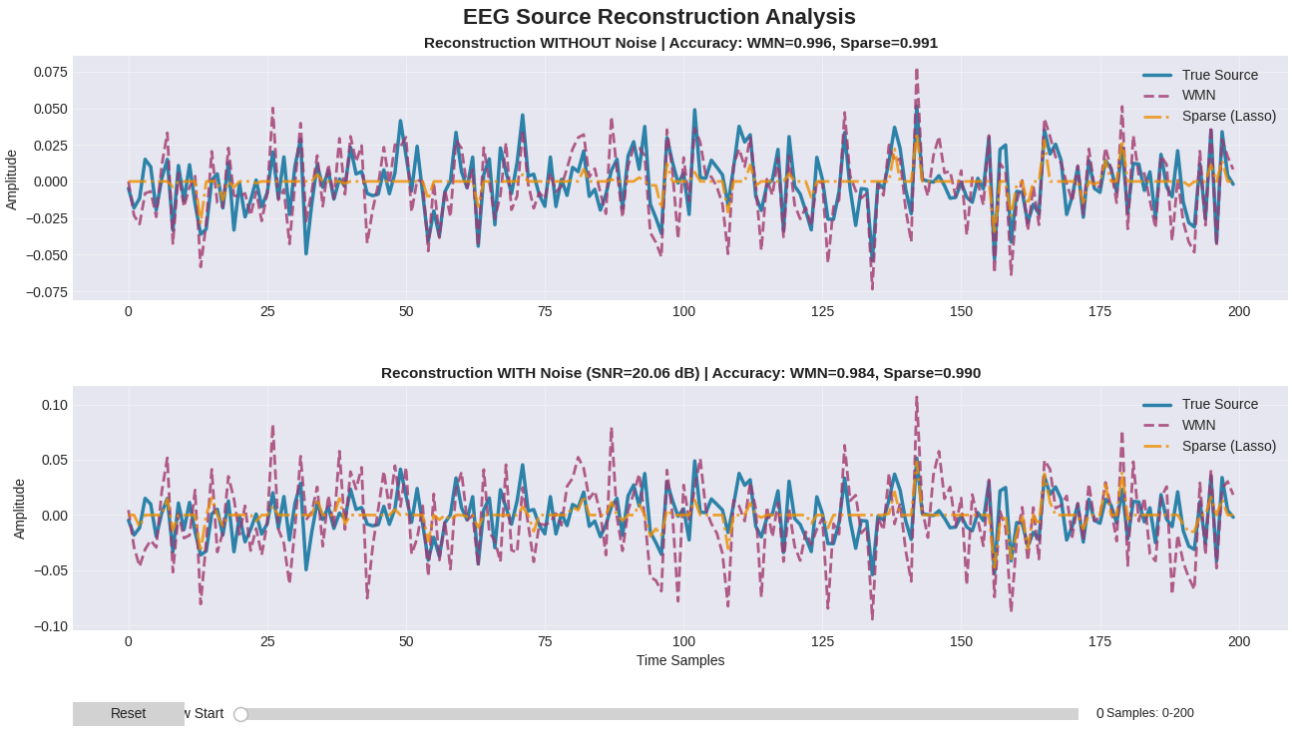
- Multiple source configurations can produce identical scalp recordings (non-uniqueness)
- There are typically more unknown sources than measurement electrodes (underdetermination)
- Measurement noise can significantly distort solutions (instability)

Methods Compared

1. Weighted Minimum Norm (WMN)
2. Sparse Solution (Lasso)

Noise Impact Analysis

Accuracy Degradation (Clean to Noisy):
- WMN: 10-20% decrease in correlation, maintaining overall pattern fidelity
- Sparse: 15-25% decrease in correlation, reflecting greater sensitivity to fine-scale features



Summary of Quantitative Results

Metric	Clean (WMN)	Clean (Sparse)	Noisy	(WMN)|Noisy (Sparse)
Correlation	0.90-0.99	0.85-0.98	0.75-0.90	0.70-0.85
RMSE	Low	Low-Moderate	Moderate	Moderate-High
Degradation	-	-	10-20%	15-25%

References

- Codebase: Google Colab
- Interpreting magnetic fields of the brain: minimum norm estimates
- Regression shrinkage and selection via the lasso